NON POWER NUCLEAR APPLICATIONS

Radioisotopes in Medicine

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- Nuclear medicine uses radiation to provide diagnostic information about the functioning of a person's specific organs, or to treat them. Diagnostic procedures using radioisotopes are now routine.
- Radiotherapy can be used to treat some medical conditions, especially cancer, using radiation to weaken or destroy particular targeted cells.
- Over 50 million nuclear medicine procedures are performed each year, and demand for radioisotopes is increasing.
- Sterilization of medical equipment is also an important use of radioisotopes.

The attributes of naturally decaying atoms, known as radioisotopes, give rise to several applications across many aspects of modern day life (see also information paper on The Many Uses of Nuclear Technology).

There is widespread awareness of the use of radiation and radioisotopes in medicine, particularly for diagnosis (identification) and therapy (treatment) of various medical conditions. In developed countries (a quarter of the world population) about one person in 50 uses diagnostic nuclear medicine each year, and the frequency of therapy with radioisotopes is about one-tenth of this.

Nuclear medicine uses radiation to provide information about the functioning of a person's specific organs, or to treat disease. In most

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cases, the information is used by physicians to make a quick diagnosis of the patient's illness. The thyroid, bones, heart, liver, and many other organs can be easily imaged, and disorders in their function revealed. In some cases radiation can be used to treat diseased organs, or tumours. Five Nobel Laureates have been closely involved with the use of radioactive tracers in medicine.

Over 10,000 hospitals worldwide use radioisotopes in medicine, and about 90% of the procedures are for diagnosis. The most common radioisotope used in diagnosis is technetium-99 (Tc-99m) accounting for about 80% of all nuclear medicine procedures and 85% of diagnostic scans in nuclear medicine worldwide.

In developed countries (about one-quarter of world population) the frequency of diagnostic nuclear medicine is about 2% per year, and the frequency of therapy with radioisotopes is about one-tenth of this. In the USA there are over 20 million nuclear medicine procedures per year, and in Europe about 10 million, with 2 million of these being therapeutic.

Nuclear medicine was developed in the 1950s by physicians with an endocrine emphasis, initially using iodine-131 to diagnose and then treat thyroid disease. In recent years specialists have also come from radiology, as dual PET/CT (positron emission tomography with computerized tomography) procedures have become established, increasing the role of accelerators in radioisotope production. However, the main radioisotopes such as Tc-99m cannot effectively be produced without reactors.*

* Some Tc-99m is produced in accelerators but it is of lower quality and at higher cost.

Nuclear medicine diagnosis, nuclear imaging

Radioisotopes are an essential part of medical diagnostic procedures. In combination with imaging devices which register the gamma rays emitted from within, they can be used for imaging to study the dynamic processes taking place in various parts of the body.

In using radiopharmaceuticals for diagnosis, a radioactive dose is given to the patient and the activity in the organ can then be studied either as a two dimensional picture or, using tomography, as a three dimensional picture. Diagnostic techniques in nuclear medicine use radioactive tracers which emit gamma rays from

within the body. These tracers are generally short-lived isotopes linked to chemical compounds which permit specific physiological processes to be scrutinized. They can be given by injection, inhalation, or orally.

The earliest technique developed uses single photons detected by a gamma camera which can view organs from many different angles. The camera builds up an image from the points from which radiation is emitted; this image is enhanced by a computer and viewed on a monitor for indications of abnormal conditions. **Single photon emission computerized tomography (SPECT)** is the current major scanning technology to diagnose and monitor a wide range of medical conditions.

A more recent development is **positron emission tomography (PET)** which is a more precise and sophisticated technique using isotopes produced in a cyclotron. A positron-emitting radionuclide is introduced, usually by injection, and accumulates in the target tissue. As it decays it emits a positron, which promptly combines with a nearby electron resulting in the simultaneous emission of two identifiable gamma rays in opposite directions. These are detected by a PET camera and give very precise indications of their origin. PET's most important clinical role is in oncology, with fluorine-18 as the tracer, since it has proven to be the most accurate non-invasive method of detecting and evaluating most cancers. It is also well used in cardiac and brain imaging.

New procedures combine PET with computed X-ray tomography (CT) scans to give co-registration of the two images (**PET-CT**), enabling 30% better diagnosis than with a traditional gamma camera alone. It is a very powerful and significant tool which provides unique information on a wide variety of diseases from dementia to cardiovascular disease and cancer.

Combining PET with MRI (**PET-MRI**), especially for brain imaging, enables diffusion-weighted imaging in soft tissue with dynamic contrast and magnetic resonance spectroscopy.

Positioning of the radiation source within (rather than external to) the body is the fundamental difference between nuclear medicine imaging and other imaging techniques such as X-rays. Gamma imaging by either method described provides a view of the position and concentration of the radioisotope within the body. Organ malfunction can be indicated if the isotope is either partially taken up in the organ (cold spot), or taken up in excess (hot spot). If a series of images is taken over a period of time, an unusual pattern

or rate of isotope movement could indicate malfunction in the organ.

A distinct advantage of nuclear imaging over X-ray techniques is that both bone and soft tissue can be imaged very successfully. This has led to its common use in developed countries where the probability of anyone having such a test is about one in two and rising.

Diagnositic radiopharmaceuticals

Every organ in our bodies acts differently from a chemical point of view. Doctors and chemists have identified a number of chemicals which are absorbed by specific organs. The thyroid, for example, takes up iodine, whilst the brain consumes quantities of glucose. With this knowledge, radiopharmacists are able to attach various radioisotopes to biologically active substances. Once a radioactive form of one of these substances enters the body, it is incorporated into the normal biological processes and excreted in the usual ways.

Diagnostic radiopharmaceuticals can be used to examine blood flow to the brain, functioning of the liver, lungs, heart, or kidneys, to assess bone growth, and to confirm other diagnostic procedures. Another important use is to predict the effects of surgery and assess changes since treatment.

The amount of the radiopharmaceutical given to a patient is just sufficient to obtain the required information before its decay. The radiation dose received is medically insignificant. The patient experiences no discomfort during the test and after a short time there is no trace that the test was ever done. The non-invasive nature of this technology, together with the ability to observe an organ functioning from outside the body, makes this technique a powerful diagnostic tool.

A radioisotope used for diagnosis must emit gamma rays of sufficient energy to escape from the body and it must have a half-life short enough for it to decay away soon after imaging is completed.

The radioisotope most widely used in medicine is Tc-99m, employed in some 80% of all nuclear medicine procedures. It is an isotope of the artificially-produced element technetium and it has almost ideal characteristics for a nuclear medicine scan, such as with SPECT. These are:

- It has a half-life of six hours which is long enough to examine metabolic processes yet short enough to minimize the radiation dose to the patient.
- It decays by an 'isomeric' process, which involves the emitting of gamma rays and low energy electrons. Since there is no high-energy beta emission the radiation dose to the patient is low.
- The low-energy gamma rays it emits easily escape the human body and are accurately detected by a gamma camera.
- The chemistry of technetium is so versatile it can form tracers by being incorporated into a range of biologically-active substances that ensure it concentrates in the tissue or organ of interest.

Its logistics also favour its use. Technetium generators – a lead pot enclosing a glass tube containing the radioisotope – are supplied to hospitals from the nuclear reactor where the isotopes are made. They contain molybdenum-99 (Mo-99), with a half-life of 66 hours, which progressively decays to Tc-99m. The Tc-99m is washed out of the lead pot by saline solution when it is required. After two weeks or less the generator is returned for recharging.

A similar generator system is used to produce rubidium-82 for PET imaging from strontium-82 – which has a half-life of 25 days.

Myocardial perfusion imaging (MPI) uses thallium-201 chloride or Tc-99m and is important for detection and prognosis of coronary artery disease.

For PET imaging, the main radiopharmaceutical is fluoro-deoxy glucose (FDG) incorporating F-18 – with a half-life of just under two hours – as a tracer. The FDG is readily incorporated into the cell without being broken down, and is a good indicator of cell metabolism.

In diagnostic medicine, there is a strong trend towards using more cyclotron-produced isotopes such as F-18, as PET and CT/PET become more widely available. However, the procedure needs to be undertaken within two hours' reach of a cyclotron, which limits their utility compared with Mo/Tc-99m.

Nuclear medicine therapy

The uses of radioisotopes in therapy are comparatively few, but nevertheless important. Cancerous growths are sensitive to damage by radiation. For this reason, some cancerous growths can be controlled or eliminated by irradiating the area containing the growth. This may be called **radiosurgery**.

External irradiation (sometimes called **teletherapy**) can be carried out using a gamma beam from a radioactive cobalt-60 source, though in developed countries the much more versatile linear accelerators are now being used as high-energy X-ray sources (gamma and X-rays are much the same). An external radiation procedure is known as gamma knife radiosurgery, and involves focusing gamma radiation from 201 sources of Co-60 on a precise area of the brain with a cancerous tumour. Worldwide, over 30,000 patients are treated annually, generally as outpatients. Teletherapy is effective in the ablation of tumours rather than their removal; it is not finely tuned.

Internal radionuclide therapy is administered by planting a small radiation source, usually a gamma or beta emitter, in the target area. Short-range radiotherapy is known as **brachytherapy**, and this is becoming the main means of treatment. Iodine-131 is commonly used to treat thyroid cancer, probably the most successful kind of cancer treatment. It is also used to treat non-malignant thyroid disorders. Iridium-192 implants are used especially in the head and breast. They are produced in wire form and are introduced through a catheter to the target area. After administering the correct dose, the implant wire is removed to shielded storage. Permanent implant seeds (40 to 100) of iodine-125 or palladium-103 are used in brachytherapy for early stage prostate cancer. Alternatively, needles with more-radioactive Ir-192 may be inserted for up to 15 minutes, two or three times. Brachytherapy procedures give less overall radiation to the body, are more localized to the target tumour, and are cost-effective.

Treating leukaemia may involve a bone marrow transplant, in which case the defective bone marrow will first be killed off with a massive (and otherwise lethal) dose of radiation before being replaced with healthy bone marrow from a donor.

Many therapeutic procedures are palliative, usually to relieve pain. For instance, strontium-89 and (increasingly) samarium-153 are used for the relief of cancer-induced bone pain. Rhenium-186 is a newer product for this.

Lutetium-177 dotatate or octreotate is used to treat tumours such as neuroendocrine ones, and is effective where other treatments

fail. A series of four treatments delivers 32 GBq. After about four to six hours, the exposure rate of the patient has fallen to less than 25 microsieverts per hour at one metre and the patients can be discharged from hospital. Lu-177 is essentially a low-energy beta-emitter (with some gamma) and the carrier attaches to the surface of the tumour.

A new field is targeted alpha therapy (TAT) or alpha radioimmunotherapy, especially for the control of dispersed (metastatic) cancers. The short range of very energetic alpha emissions in tissue means that a large fraction of that radiative energy goes into the targeted cancer cells, once a carrier such as a monoclonal antibody has taken the alpha-emitting radionuclide such as bismuth-213 to the areas of concern. Clinical trials for leukaemia, cystic glioma, and melanoma are underway. TAT using lead-212 is increasingly important for treating pancreatic, ovarian, and melanoma cancers.

An experimental development of this is Neutron Capture Enhanced Particle Therapy (NCEPT) which involves injecting a patient with a neutron capture agent shortly before irradiation with protons or heavy ions. This approach boosts the target dose without increasing the dose to healthy tissue and delivers a significant dose to secondary lesions outside the primary treatment area. It uses boron-10 or gadolinium-157 which concentrate in malignant brain tumours. The patient is then irradiated with thermal neutrons or protons which are strongly absorbed by the boron, producing high-energy alpha particles which kill the cancer. This requires the patient to be brought to a nuclear reactor, rather than the radioisotopes being taken to the patient.

Radionuclide therapy has progressively become more successful in treating persistent disease and doing so with low toxic side-effects. With any therapeutic procedure the aim is to confine the radiation to well-defined target volumes of the patient. The doses per therapeutic procedure are typically 20-60 Gy.

Treatment may involve significant radioactivity (e.g. 4.4 GBq is quoted as an average dose of I-131 for thyroid ablation, and up to 11 GBq for patients with advanced metastatic disease). According to US regulatory guidelines for I-131, the patient can be released if the activity is below 1.2 GBq, or 0.07 mSv/hr at 1 metre. Meanwhile a lot of I-131 is flushed down the hospital toilet and plumbing needs to be shielded accordingly.

Therapeutic radiopharmaceuticals

For some medical conditions, it is useful to destroy or weaken malfunctioning cells using radiation. The radioisotope that generates the radiation can be localized in the required organ in the same way it is used for diagnosis – through a radioactive element following its usual biological path, or through the element being attached to a suitable biological compound. In most cases, it is beta radiation which causes the destruction of the damaged cells. This is radionuclide therapy (RNT) or radiotherapy. Short-range radiotherapy is known as brachytherapy, and this is becoming the main means of treatment.

Although radiotherapy is less common than diagnostic use of radioactive material in medicine, it is nevertheless widespread, important, and growing. An ideal therapeutic radioisotope is a strong beta emitter with just enough gamma to enable imaging (e.g. lutetium-177). This is prepared from ytterbium-176 which is irradiated to become Yb-177 (which decays rapidly to Lu-177). Yttrium-90 is used for treatment of cancer, particularly non-Hodgkin's lymphoma and liver cancer, and it is being used more widely, including for arthritis treatment. Lu-177 and Y-90 are becoming the main RNT agents.

Iodine-131, samarium-153, and phosphorus-32 are also used for therapy. I-131 is used to treat the thyroid for cancers and other abnormal conditions such as hyperthyroidism (over-active thyroid). In a disease called Polycythemia vera, an excess of red blood cells is produced in the bone marrow. P-32 is used to control this excess.

Caesium-131, palladium-103, and radium-223 are also used for brachytherapy, all being Auger (soft) X-ray emitters, and having half-lives of 9.7 days, 17 days, and 11.4 days, respectively, much less than the 60 days of I-125 which they replace.

A new and still experimental procedure uses boron-10, which concentrates in the tumour. The patient is then irradiated with neutrons which are strongly absorbed by the boron, to produce high-energy alpha particles which kill the cancer. This is boron neutron capture therapy.

For targeted alpha therapy (TAT), actinium-225 is used, from which the daughter bismuth-213 can be obtained (via three alpha decays) to label targeting molecules. The bismuth is obtained by elution from an Ac-225/Bi-213 generator similar to the Mo-99/Tc-99m one. Bi-213 has a 46-minute half-life. The Ac-225 (half-life 10 days) is

formed from radioactive decay of radium-225, the decay product of long-lived thorium-229, which is obtained from decay of uranium-233, which in turn is formed from thorium-232 by neutron capture in a nuclear reactor. A Th-229/Ac-225 generator is eluted about eight times per year at ORNL, but output is limited. Ac-225 itself is an alpha-emitter and may be used directly, bonded to a protein or antibody such as PSMA for prostate cancer. Larger quantities of Ac-225 are being produced in Canada by TRIUMF with a high-energy proton beam on a Th-232 target. Production of Ac-225 has expanded significantly: TerraPower Isotopes began commercial-scale production in October 2024; Cardinal Health started weekly production in December 2024; and NorthStar Medical Radioisotopes began routine weekly Ac-225 production in late 2025.

Another radionuclide recovered from Th-232, but by natural decay via thorium-228, is Pb-212, with a half-life of 10.6 hours. Pb-212 can be attached to monoclonal antibodies for cancer treatment by TAT. A Ra-224/Pb-212 generator system similar to the Mo-99/Tc-99m one is used to provide Pb-212 from Ra-224 (via Ra-220 and polonium-216 (po-216)). Pb-212 has a half-life of 10.6 hours, and beta decays to Bi-212 (1 hour half-life), then most beta decays to Po-212. The alpha decays of Bi-212 and Po-212 are the active ones destroying cancer cells over a couple of hours. Stable Pb-208 results, via Tl-208 for the bismuth decay.

TAT also uses Ra-223, Th-227, Ac-225 and astatine-211.

Considerable medical research is being conducted worldwide into the use of radionuclides attached to highly specific biological chemicals such as immunoglobulin molecules (monoclonal antibodies). The eventual tagging of these cells with a therapeutic dose of radiation may lead to the regression – or even cure – of some diseases.

Sterilization

Many medical products today are sterilized by gamma rays from a Co-60 source, a technique which generally is much cheaper and more effective than steam heat sterilization. The disposable syringe is an example of a product sterilized by gamma rays. Because it is a 'cold' process radiation can be used to sterilize a range of heat-sensitive items such as powders, ointments, and solutions, as well as biological preparations such as bone, nerve, and skin to be used in tissue grafts. Large-scale irradiation facilities for gamma

sterilization are installed in many countries. Smaller gamma irradiators, often utilising Cs-137, having a longer half-life, are used for treating blood for transfusions and for other medical applications.

Sterilization by radiation has several benefits. It is safer and cheaper because it can be done after the item is packaged. The sterile shelf-life of the item is then practically indefinite provided the seal is not broken. Irradiation technologies are used to sterlize almost half of the global supply of single-use medical products. Apart from syringes, medical products sterilized by radiation include cotton wool, burn dressings, surgical gloves, heart valves, bandages, plastic, and rubber sheets and surgical instruments.

Supply of radioisotopes

The main world isotope suppliers are Curium (France & USA), MDS Nordion (Canada), IRE (Europe), NTP (South Africa), JSC Isotope (Russia), and ANM (ANSTO Australia).

Most medical radioisotopes made in nuclear reactors are sourced from relatively few research reactors, including:

- HFR at Petten in Netherlands (supplied via IRE and Curium).
- BR-2 at Mol in Belgium (supplied via IRE and Curium).
- Maria in Poland (supplied via Curium).
- Orphee at Saclay in France (supplied via IRE).
- FRJ-2/FRM-2 at Jülich in Germany (supplied via IRE).
- LVR-15 at Rez in Czech Republic (supplied via IRE).
- HFETR at Chengdu in China.
- Safari in South Africa (supplied from NTP).
- OPAL in Australia (supplied from ANM).
- ETRR-2 in Egypt (forthcoming: supplied to domestic market).
- Dimitrovgrad in Russia (supplied via JSC Isotope).

- NRU at Chalk River in Canada (supplied via MDS Nordion) ceased production in October 2016, and the reactor closed down in March 2018.

Of fission radioisotopes, the vast majority of demand is for Mo-99 (for Tc-99m).

Supply capacity is always substantially (e.g. 50%) above demand, due to decay of Mo-99 in transit, despite the six-day TBq/Ci quantification. One challenge is the delivery of fresh supplies in weekdays, in line with demand, to minimize waste.

Russia is keen to increase its share of world supply, and in 2017 JSC Rusatom Healthcare was set up by Atomenergoprom to integrate Russian radioisotope efforts, including JSC Isotope. It is for production and distribution of medical isotopes and radiopharmaceuticals with Rosenergoatom (about 80% of the Russian market), and for the ionizing treatment of medical products, mainly Co-60 produced in power reactors. In 2018 it was planning to build a RUR 6 billion radiopharmaceutical plant near Moscow. In January 2023 Russia began constructing a medical isotope plant in Obninsk to manufacture products based on iodine-131, samarium-153 and molybdenum-99 as well as active radiopharmaceutical drugs based on lutetium-177, actinium-225 and other isotopes. Rosatom plans to have production lines operating by 2025. Russia has limited spare capacity for export.

World demand for Mo-99 was 23,000 six-day TBq/yr* in 2012, but has apparently dropped back to about 18,500 since. Mo-99 is mostly produced by fission of U-235 targets in a nuclear research reactor. The targets are then processed to separate the Mo-99 and also to recover I-131. As of March 2023 all reactors producing Mo-99 use low-enriched uranium (LEU) targets instead of proliferation-sensitive highly enriched uranium (HEU), which adds about 20% to production costs. However, in medical imaging, the cost of Mo-99 itself is small relative to hospital costs. Mo-99 can also be made by bombarding Mo-98 with neutrons in a reactor. However, this activation Mo-99 has relatively low specific activity, with a maximum of 74 GBq/g (depending on the neutron flux available in the reactor), compared with 185 TBq/g or more for conventional fission-produced Mo-99.

* 23,000 TBq is on the basis of activity at 6 days from production reference point, *i.e.* 22% of nearly 100,000 TBq required in production processing (given 66 hour half-life). This is still about two days from the end of irradiation, so some 167,000 TBq/yr must

be made in the actual reactors to allow for cooling, processing and decay en route to the users.

There are three ways to produce Mo-99. The most common and effective method is by fission of uranium in a target foil, followed by chemical separation of the Mo. This fission is done in research reactors. A second method is neutron activation, where Mo-98 in target material captures a neutron. This is done in power reactors, usually RBMK or Candu. A third method is by proton bombardment of Mo-100 in an accelerator of some kind. There are plans to produce it by fission in a subcritical assembly in an accelerator.

Threatened supply constraints

A number of incidents in 2008 pointed out shortcomings and unreliability in the supply of medical isotopes, particular technetium. In September 2008 the World Council on Isotopes was set up, based in South Korea, to promote isotope technologies.

As indicated above, the world's supply of Mo-99 comes from just six reactors, five of which are over 50 years old. In September 2025, construction began on the PALLAS reactor in Petten, Netherlands, which will replace the High Flux Reactor (HFR) around 2030. The Canadian and Netherlands reactors required major repairs over 2009-10 and were out of action for some time. Osiris was due to shut down in 2015 but apparently continued to at least 2016. NRU at Chalk River was re-licensed to October 2016 when it ceased production, and was fully retired in March 2018. A new 15 MW South Korean reactor at Busan – KJRR – is under construction. An increasing supply shortfall of technetium-99 was forecast from 2010, and the IAEA encouraged new producers. Also, the processing and distribution of isotopes is complex and constrained, which can be critical when the isotopes concerned are short-lived. A need for increased production capacity and more reliable distribution is evident. The Mo-99 market is about $5 billion per year, according to NECSA.

In 2009 the NEA set up the High-level Group on the Security of Supply of Medical Radioisotopes (HLG-MR) to strengthen the reliability of Mo-99 and Tc-99m supply in the short, medium, and long term. It reviewed the Mo-99 supply chain to identify the key areas of vulnerability, the issues that need to be addressed, and the mechanisms that could be used to help resolve them. It requested an economic study of the supply chain, and this was published in 2010 by the NEA. The report identifies possible changes needed.

The NEA report predicted supply shortages from 2016, not simply from reactors but due to processing limitations too. Historically reactor irradiation prices have been too low to attract new investment, and full cost recovery is needed to encourage new infrastructure. This will have little impact on end prices since irradiation only accounts for about 1% of product cost. Transport regulation and denial of shipment impede reliable supply. HEU use needs to be minimized, though conversion to LEU targets will reduce capacity. Outage reserve capacity needs to be sourced, valued, and paid for by the supply chain. Fission is the most efficient and reliable means of production, but Canada and Japan are developing better accelerator-based techniques.

The supply situation led, in December 2014, to the NEA Joint Declaration on the Security of Supply of Medical Radioisotopes, focused on Mo-99, supported by 13 countries: Australia, Canada, France, Germany, Japan, the Netherlands, Poland, South Korea, Russia, South Africa, Spain, the UK, and the USA. A review of the situation in mid-2017 showed that the market had substantially restructured following the 2009-2010 supply crisis, and that restructuring had led to increased efficiencies in the use of material at the different layers in the supply chain. The latest NEA data confirms a relatively flat market demand of around 333 six-day TBq Mo-99 per week at the end of radiochemical processing. In addition, several sources of supply had ramped up production to lift the baseline supply capacity for the 2017 and 2018 periods to a level safely above the revised market demand.

US supply initiatives for molybdenum-99

The US Congress has called for all Mo-99 to be supplied by reactors running on low-enriched uranium (LEU), instead of high-enriched uranium (HEU). Also it called for proposals for an LEU-based supply of Mo-99 for the US market, reaching 111 six-day TBq per week by mid-2013, a quarter of world demand. Tenders for this closed in June 2010, but evidently no immediate progress was made. In December 2012 Congress passed the American Medical Isotope Production Act of 2011 to establish a technology-neutral program to support the production of Mo-99 for medical uses in the USA by non-federal entities.

In February 2019, the Department of Energy's National Nuclear Security Administration (NNSA) selected four companies to begin negotiations for potential new cooperative agreement awards for the supply of molybdenum-99, mostly from accelerators. Niowave

is developing superconducting electron linear accelerators, NorthStar Medical Radioisotopes is planning to irradiate Mo-98 targets to produce Mo-99 in a reactor, while in the longer term it is developing a method using a linear accelerator. Northwest Medical Isotopes (NWMI) plans to use LEU targets in a Triga research reactor, and Shine Medical Isotopes is building a plant to use a low-energy accelerator-based neutron source to fission an LEU target in aqueous solution. All were awarded cooperative agreements of $15 million each by August 2019 for Mo-99 production, which the NNSA said should total about 110 six-day TBq per week. See below for fuller descriptions.

NorthStar Medical Radioisotopes formerly used the University of Missouri research reactor (MURR) to produce Mo-99 via neutron activation of Mo-98 targets, separated using its RadioGenix system. However, NorthStar discontinued Mo-99 production at the end of 2023 to focus on therapeutic isotopes including actinium-225 and copper-67.

In 2014 another plan using the US University Reactor Network was announced. Northwest Medical Isotopes (NWMI) planned to produce half of North America's demand for Mo-99 from 2017, using LEU targets, with licensed technology from Oregon State University. In May 2018 NWMI received NRC approval to build its production facility at Columbia, Missouri; however, in February 2022 NWMI requested termination of its construction permit, having never commenced construction.

In February 2015, Nordion and its US parent Sterigenics International announced a new arrangement with the University of Missouri research reactor (MURR) and General Atomics to produce Mo-99 from LEU targets from 2018 using the 10 MW pool-type reactor. By December 2016 the project was funded to $25 million by NNSA. The new medical isotope supply was to be produced using General Atomics' innovative Selective Gaseous Extraction (SGE) technology to extract the molybdenum from the targets. A licence application was submitted to the NRC in March 2017, with a view to meeting half of US demand for Mo-99. However, Nordion withdrew from the project in April 2018 citing delays and cost overruns that had increased the project's commercial risk.

In the USA Coquí Pharmaceuticals has signed a contract with Argentinian nuclear engineering company INVAP to build an open-pool reactor similar to Australia's Opal, using LEU targets, and a Mo-99 production facility at Oak Ridge.

An earlier proposal for Mo-99 production involving an innovative reactor and separation technology has lapsed. In January 2009 Babcock & Wilcox (B&W) announced an agreement with international isotope supplier Covidien to produce Mo-99 sufficient for half of US demand, if a new process was successful. They planned to use Aqueous Homogeneous Reactor (AHR) technology with LEU in small 100-200 kW units where the fuel is mixed with the moderator and the U-235 forms both the fuel and the irradiation target.* A single production facility could have four such reactors. B&W and Covidien expected a five-year lead time to first production. B&W received $9 million towards this Medical Isotope Production System (MIPS) in 2010 from the US government and completed the R&D and conceptual design phase in 2012. However, in October 2012 Covidien pulled out of the joint venture with B&W "after learning that the time and cost involved with the project would be greater than originally expected." Covidien said that it was "making significant long-term capital investment in a new Tc-99m generator facility at our US plant, and conversion from HEU- to LEU-based Mo-99 production at our processing plant in the Netherlands." B&W appears to have dropped the MIPS.

* LEU is dissolved in acid then brought to criticality in a 200-litre vessel. As fission proceeds the solution is circulated through an extraction facility to remove the fission products with Mo-99 and then back into the reactor vessel, which is at low temperature and pressure.

In mid-2013 Los Alamos National Laboratory announced that it had recovered Mo-99 from low-enriched sulphate reactor fuel in solution, raising the prospect of this process becoming associated with commercial reprocessing plants as at La Hague in France.

Russian supply initiatives for Mo-99

In Russia, the Research Institute of Atomic Reactors (NIIAR or RIAR, with three reactors for isotope production) and Trans-regional Izotop Association (becoming JSC Isotope in 2008) established a joint venture, Isotop-NIIAR, to produce Mo-99 at Dimitrovgrad from 2010. Phase 1 of the Mo-99 production complex with capacity of 1700 TBq/yr was commissioned in December 2010, and Phase 2 was commissioned in June 2012 taking total capacity to 1480 TBq/yr (evidently 6-day activity). Earlier reports quoted 4800 TBq/yr, and Rosatom aimed for 20% of the world Mo-99 market by 2014, supplied internationally through Nordion. In September 2010 JSC Isotope signed a framework agreement with MDS Nordion to

explore commercial opportunities outside Russia on the basis of this Isotop-NIIAR JV, initially over ten years.

JSC Isotope was founded in 1958 and incorporated in 2008. In 2010-11 it set up Mo-99 production at JSC RIAR Dimitrovgrad and has since expanded and diversified. It connects Russian isotope manufacturers with domestic and foreign marketplaces, under JSC Rusatom Healthcare. Brazil is a major export market. One production facility is the Obninsk branch of the Karpov Institute of Physcal Chemistry. Its product portfolio includes more than 60 radioisotopes produced in cyclotrons, nuclear reactors by irradiation of targets, or recovered from spent nuclear fuel, as well as hundreds of types of ionizing radiation sources and compounds tagged with radioactive isotopes. It has more than 10,000 scientific and industrial customers for industrial isotopes in Russia. The Karpov Institute gets some supply from Leningrad nuclear power plant.

At Russia's Kurchatov Institute the 20 kW ARGUS Aqueous Homogeneous Reactor (AHR) has operated since 1981, and R&D on producing Mo-99 from it is ongoing.

Other reactor supply initiatives for Mo-99

Australia's Opal reactor has the capacity to produce half the world supply of Mo-99, and with the ANSTO Nuclear Medicine Project supplies about one-quarter of world demand. ANSTO is building a substantial Mo-99 production facility to ramp up quickly to 130 six-day TBq per week (6500 per year), or 10 million Tc-99m doses per year, with exports to the USA, Japan, China, and Korea. ANSTO increased production from 30 to 80 six-day TBq/wk (1500 to 4000 per year) from July 2016.

In the USA, Lantheus Medical Imaging (LMI) manufactures Tc-99m generators from Mo-99 produced by ANSTO in Australia, NTP in South Africa and IRE in Belgium (drawing on BR-2 in Belgium, HFR in Netherlands and LVR-15 in Czech Republic). In 2014 it signed an agreement with Shine Medical Technologies in the USA for future supplies of Mo-99.

During the 2009-10 supply crisis, NECSA's Safari reactor (in South Africa) was able to supply over 25% of the world's Mo-99.

Areva and Bruce Power are aiming to produce Mo-99 among other isotopes in the latter's Candu power reactors – see below.

In 2024, Laurentis Energy Partners and BWXT Medical began irradiating and harvesting Mo-99 at Darlington. Darlington is the

world's first commercial power reactor to produce Mo-99.

Non-reactor technetium

Tc-99m or Mo-99 can also be produced in small quantities from cyclotrons and accelerators, in a cyclotron by bombarding a Mo-100 target with a proton beam to produce Tc-99m directly, or in a linear accelerator to generate Mo-99 by bombarding an Mo-100 target with high-energy X-rays. It is generally considered that non-reactor methods of producing large quantities of useful Tc-99m are some years away. At present the cost is at least three times and up to ten times that of the reactor route, and Mo-100 is available only from Russia. If Tc-99m is produced directly in a cyclotron, it needs to be used quickly, and the co-product isotopes are a problem.

SHINE Medical Technologies and Phoenix are developing an advanced accelerator technology for the production of Mo-99 as a fission product. SHINE has been awarded $114 million in grants from the NNSA to January 2026, and it has a $125 million debt financing package from healthcare investment firm Deerfield Management. An LEU target solution is irradiated with low-energy neutrons in a subcritical assembly – not a nuclear reactor. The neutrons are generated through a beam-target fusion reaction caused by accelerating deuterium ions into tritium gas, using a particle accelerator. SHINE is an acronym for 'subcritical hybrid intense neutron emitter'.

In June 2016 China's largest producer and distributor of medical radioisotopes, HTA, entered a strategic agreement for the supply of SHINE's Mo-99. Construction at Janesville, Wisconsin commenced in August 2017 on 'Building One' and in May 2019 on the main production facility, which would eventually be capable of producing nearly half of global Mo-99 demand when fully operational. A 132-hour test run of Phoenix's high-flux neutron generator was in June 2019. Eight of these will be installed by SHINE.

In June 2021 SHINE announced a contract with University of Missouri Research Reactor (MURR) for irradiation of ytterbium-176 to produces lutetium-177. Its Cassiopeia plant at Janesville is the largest producer of Lu-177 in the USA. EDH Nuclear Medicine distributes the Lu-177.

A year later in Canada, Lu-177 was produced using a new isotope production system, which was installed in unit 7 of the Bruce plant in Ontario during a maintenance outage. The process included the irradiation of ytterbium-176 to produce Lu-177, which was then

sent for processing in Germany where Isotope Technologies Munich isolated pharmaceutical-grade Lu-177. Furthermore, the Canadian government has an Isotope Technology Acceleration Program (ITAP) to promote R&D on non-reactor based isotope production, particularly through the Medical Isotope Program (MIP). Canada Light Source (CLS) in Saskatoon is using a linear accelerator to bombard Mo-100 targets with X-rays, and has produced some Mo-99 for MIP.

At Lansing in Michigan, Niowave is using a superconducting electron linear accelerator to produce isotopes from fission of low-enriched uranium. It reports production of Mo-99, I-131, Sr-89 and Xe-133 among others.

NorthStar planned to produce Mo-99 from Mo-100 in an accelerator, and in December 2016 received $11 million from NNSA for this. However, at the end of 2023 NorthStar announced it was shutting down its Mo-99 production facilities in Beloit, Wisconsin, citing increasing costs and competition. The company has since changed its focus on producing actinium-225 and copper-67 for cancer therapy using its electron beam accelerators.

Main Mo-99 production reactors

	Reactor	Targets	Capacity*	Start	Est. stop
Belgium	BR-2	HEU	6500	1961	2026
Netherlands	HFR	LEU**	6200	1961	2030
Australia	OPAL	LEU	3500	2006	2030+
Czech Republic	LVR-15	HEU	3000	1989	2028
South Africa	Safari-1	LEU	3000	1965	2030
Poland	Maria	LEU	2200	1974	2030
Russia	RIAR: three	HEU	890	1961-70	
USA	MURR	HEU	750	1966	
Argentina	RA-3	LEU	400	1967	2027
Total			26,440		

* Six-day Ci ^{99}Mo
** In March 2021 HFR started using solely LEU for production of radioisotopes for medical use.

Source: OECD-NEA, The Supply of Medical Radioisotopes: An Economic Diagnosis and Possible Solutions (2019)

Other medical radioisotopes

Cobalt-60 has mostly come from Candu power reactors by irradiation of Co-59 in special rods for up to three years (or five in RBMK), and production is being expanded. Production sites include: Bruce B and Pickering in Canada (expanding to Bruce A and Darlington); Embalse in Argentina; Qinshan Phase III units 1&2 in China; Wolsong 1&2 in South Korea (all Candu); and Leningrad 3&4, Smolensk 1-3 and Kursk 4 (all RBMK, where the Co-59 replaces dysprosium absorbers), and Beloyarsk 4 (BN-800 fast neutron reactor) in Russia.

Most of this Co-60 is used for sterilization, with high-specific-activity Co-60 for cancer treatment. Formerly this radioisotope was made in Canada's NRU at Chalk River until it closed but Bruce B nuclear power plant has increased its output for Co-60, and in February 2025 completed its largest harvest of cobalt-60 in the company's history. Rosenergoatom in Russia is implementing plans to increase its global market share of Co-60 from 15% to 30%, mainly from Kursk. Much of the Co-60 is supplied through Nordion.

Under an August 2017 agreement between Areva NP and Bruce Power, Areva will design and supply equipment to be installed in the existing Bruce Candu units to add online production at commercial scale of "a wide range of isotopes for use in both health care and industry." In particular, this will enable the plant to produce short half-life isotopes such as Mo-99, lutetium-177 and iridium-192 using a system that inserts and removes targets with little impact on the normal operation of the power reactors. The process will use Areva NP's patent-pending method of producing radioisotopes using a heavy water nuclear power plant.

Orano Med built a small plant at Bessines-sur-Gartempe in France to provide Pb-212 from irradiated thorium, and this came online in 2013. It was extended with a fivefold increase in capacity in 2021. A second plant has been built at Plano in Texas, operating from 2016, and a new industrial-scale plant is planned for Caen in France. A radium-224/Pb-212 generator similar to the Mo-99/Tc-99m one enables the Pb-212 to be eluted as required for targeted alpha therapy (TAT). Ra-224 is a natural decay product of Th-228, and indirectly, of Th-232.

Some iodine-131 is produced at Leningrad nuclear power plant from tellurium oxide, using irradiation channels in the RBMK reactors. A contract with the Karpov Institute of Physical Chemistry provides for delivery of 2.6-3.0 TBq of I-131 per week. The plant

also produces Co-60, I-125, and Mo-99 for Karpov IPC. In 2017 Rosatom announced the establishment of a radiopharmaceutical production plant at the Institute of Reactor Materials (IRM), which had started with lutetium-177, producing 24 TBq of it in 2016 (650 Ci). The IRM will also produce iodine-125 and iridium-192, and its products will be distributed through Isotop.

Urenco Stable Isotopes at Almelo uses centrifuge technology to produce by centrifuge enrichment a variety of stable isotopes for medical applications. A new cascade commissioned in 2021 is designed to produce multiple isotopes, including those of cadmium, germanium, iridium, molybdenum, selenium, tellurium, titanium, tungsten, xenon and zinc.

Isotopes used in medicine

Many radioisotopes are made in nuclear reactors, some in cyclotrons. Generally neutron-rich ones and those resulting from nuclear fission need to be made in reactors; neutron-depleted ones such as PET radionuclides are made in cyclotrons with energy ranging from 9 to 19 MeV. Higher-energy machines of about 30 MeV are needed for most SPECT radionuclides. There are about 40 activation product radioisotopes and five fission product ones made in reactors.

Reactor radioisotopes

Bismuth-213 (half-life: 46 min):
Used for targeted alpha therapy (TAT), especially cancers, as it has a high energy (8.4 MeV).

Caesium-131 (9.7 d):
Used for brachytherapy, emits soft X-rays.

Caesium-137 (30 yr):
Used for low-intensity sterilization of blood and in brachytherapy.

Chromium-51 (28 d):
Used to label red blood cells for monitoring, and to quantify gastro-intestinal protein loss or bleeding.

Cobalt-60 (5.27 yr):
Formerly used for external beam radiotherapy, now almost universally used for sterilising. High-specific-activity (HSA) Co-60 is used for brain cancer treatment.

Dysprosium-165 (2 h):
Used as an aggregated hydroxide for synovectomy treatment of arthritis.

Erbium-169 (9.4 d):
Used for relieving arthritis pain in synovial joints.

Holmium-166 (26 h):
Being developed for diagnosis and treatment of liver tumours. Administered as microspheres.

Iodine-125 (60 d):
Used in cancer brachytherapy (prostate and brain), also diagnostically to evaluate the filtration rate of kidneys and to diagnose deep vein thrombosis in the leg. It is also widely used in radioimmuno-assays to show the presence of hormones in tiny quantities.

Iodine-131 (8 d)*:
Widely used in treating thyroid cancer and in imaging the thyroid; also in diagnosis of abnormal liver function, renal (kidney) blood flow, and urinary tract obstruction. A strong gamma emitter, but used for beta therapy.

Iridium-192 (74 d): Supplied in wire form for use as an internal radiotherapy source for cancer treatment (used then removed), *e.g.* for prostate cancer. Strong beta emitter for high dose-rate brachytherapy.

Iron-59 (46 d):
Used in studies of iron metabolism in the spleen.

Lead-212 (10.6 h):
Used in TAT for cancers or alpha radioimmunotherapy, with decay products Bi-212 (1 h) and Po-212 (0.3 μs) delivering the alpha particles. Used especially for melanoma, breast cancer and ovarian cancer. Demand is increasing. Used in peptide receptor radionuclide therapy (PRRT).

Lutetium-177 (6.7 d):
Lu-177 is increasingly important as it emits just enough gamma for imaging while the beta radiation does the therapy on small (eg endocrine) tumours. Its half-life is long enough to allow sophisticated preparation for use. It is usually produced by neutron activation of natural or enriched lutetium-176 targets or indirectly by neutron irradiation of Yb-176.

Molybdenum-99 (66 h)*:
Used as the 'parent' in a generator to produce technetium-99m.

Palladium-103 (17 d):
Used to make brachytherapy permanent implant seeds for early stage prostate cancer. Emits soft x-rays.

Phosphorus-32 (14 d):
Used in the treatment of polycythemia vera (excess red blood cells). Beta emitter.

Potassium-42 (12 h):
Used for the determination of exchangeable potassium in coronary blood flow.

Radium-223 (11.4 d):
Used for TAT brachytherapy, lodges in bone, emits soft X-rays.

Rhenium-186 (3.8 d):
Used for pain relief in bone cancer. Beta emitter with weak gamma for imaging.

Rhenium-188 (17 h):
Used to beta irradiate coronary arteries from an angioplasty balloon.

Samarium-153 (47 h):
Sm-153 is very effective in relieving the pain of secondary cancers lodged in the bone, sold as Quadramet. Also very effective for prostate and breast cancer. Beta emitter.

Scandium-47 (4.5 d)
Sc-47 has similar properties to Lu-177 and may be used for therapy or diagnosis. It is produced by irradiating calcium-46 to produce Ca-47 which decays to Sc-47.

Selenium-75 (120 d):
Used in the form of seleno-methionine to study the production of digestive enzymes.

Sodium-24 (15 h):
For studies of electrolytes within the body.

Strontium-89 (50 d)*:
Very effective in reducing the pain of prostate and bone cancer. Beta emitter.

Technetium-99m (6 h):
Used in to image the skeleton and heart muscle in particular, but also for brain, thyroid, lungs (perfusion and ventilation), liver, spleen, kidney (structure and filtration rate), gall bladder, bone marrow, salivary and lacrimal glands, heart blood pool, infection, and numerous specialized medical studies. Produced from Mo-99 in a

generator. The most common radioisotope for diagnosis, accounting for over 80% of scans.

Thorium-227 (18.7 d):
Used for TAT, decays to Ra-223.

Xenon-133 (5 d)*:
Used for pulmonary (lung) ventilation studies.

Ytterbium-169 (32 d):
Used for cerebrospinal fluid studies in the brain.

Ytterbium-177 (1.9 h):
Progenitor of Lu-177, through neutron irradiation of Yb-176.

Yttrium-90 (64 h)*:
Used for cancer brachytherapy and as silicate colloid for the relieving the pain of arthritis in larger synovial joints. Pure beta emitter and of growing significance in therapy, especially liver cancer.

Radioisotopes of gold and ruthenium are also used in brachytherapy.

* fission product

Cyclotron radioisotopes

Actinium-225 (10 d):
Used for TAT especially prostate cancers.

Astatine-211 (7.2 h):
Use for TAT.

Bismuth-213 (46 min):
Used for TAT.

Carbon-11, Nitrogen-13, Oxygen-15, Fluorine-18:
These are positron emitters used in PET for studying brain physiology and pathology, in particular for localizing epileptic focus, and in dementia, psychiatry, and neuropharmacology studies. They also have a significant role in cardiology. F-18 in FDG (fluorodeoxyglucose) has become very important in detection of cancers and the monitoring of progress in their treatment, using PET.

Cobalt-57 (272 d):
Used as a marker to estimate organ size and for in-vitro diagnostic kits.

Copper-64 (13 h):
Used to study genetic diseases affecting copper metabolism, such as Wilson's and Menke's diseases, for PET imaging of tumours, and also cancer therapy.

Copper-67 (2.6 d):
Beta emitter, used in therapy.

Fluorine-18 (110 min) as FLT (fluorothymidine), F-miso (fluoromisonidazole), 18F-choline:
It decays with positron emission, so used as tracer with PET, for imaging malignant tumours.

Gallium-67 (78 h):
Used for tumour imaging and locating inflammatory lesions (infections).

Gallium-68 (68 min):
Positron emitter used in PET and PET-CT units. Derived from germanium-68 in a generator.

Germanium-68 (271 d):
Used as the 'parent' in a generator to produce Ga-68.

Indium-111 (2.8 d):
Used for specialist diagnostic studies, e.g. brain studies, infection and colon transit studies. Also for locating blood clots, inflammation and rare cancers.

Iodine-123 (13 h):
Increasingly used for diagnosis of thyroid function, it is a gamma emitter without the beta radiation of I-131.

Iodine-124 (4.2 d):
Tracer, with longer life than F-18, one-quarter of decays are positron emission so used with PET. Also used to image the thyroid using PET.

Krypton-81m (13 sec) from rubidium-81 (4.6 h):
Kr-81m gas can yield functional images of pulmonary ventilation, e.g. in asthmatic patients, and for the early diagnosis of lung diseases and function.

Rubidium-82 (1.26 min):
Convenient PET agent in myocardial perfusion imaging.

Strontium-82 (25 d):
Used as the 'parent' in a generator to produce Rb-82.

Thallium-201 (73 h):
Used for diagnosis of coronary artery disease other heart conditions such as heart muscle death and for location of low-grade lymphomas. It is the most commonly used substitute for technetium-99 in cardiac-stress tests.

Notes & references

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Related information

Radioisotopes in Industry
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